

ATIS-0100072

Network Reliability Steering Committee 2023-2024
Operational Report



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Network Reliability Steering Committee 2023-2024 Operational Report

June 2025

DATE: June 2025

TO: Stakeholders of the Nation's Public Communications Networks

Communications networks play a critical role in supporting the daily lives of individuals and businesses across the nation, underscoring their importance in both routine activities and during infrequent service disruptions. In times of disruption, communication providers consistently exemplify their steadfast dedication to ensuring reliable services. Their commitment is evident in the extraordinary resources they deploy to swiftly restore services, analyze disruptions, and implement meaningful improvements. Network owners, operators, and their equipment partners uphold an enduring pledge to build and maintain resilient and reliable networks—a commitment that serves as the foundation of our modern society and shines during both ordinary operations and moments of crisis.

At the forefront of these efforts stands the Alliance for Telecommunications Industry Solutions (ATIS) Network Reliability Steering Committee (NRSC), a collaborative industry body that works tirelessly to strengthen the nation's communications infrastructure. By analyzing outage trends and addressing complex regulatory and technical challenges, the NRSC ensures our networks remain secure and dependable, benefiting consumers, businesses, and the nation as a whole. This partnership with industry leaders has been instrumental in navigating pivotal initiatives, such as the implementation of the FCC's 988 Suicide Lifeline requirements. By engaging stakeholders, the NRSC has facilitated smoother regulatory processes, ultimately reducing burdens and ensuring public safety.

This report reflects the NRSC's remarkable achievements over the past two years (2023–2024). Its work has focused on developing a cohesive approach to collecting and analyzing meaningful data, enabling a deeper understanding of outage causes and fostering improved situational awareness of the nation's communications health. In addition, the NRSC continues to champion Industry Best Practices, ensuring they remain current and relevant in a rapidly evolving technological landscape. As the nation continues to transition to a more interconnected world with expanded wireless networks, the rise of 5G, and the shift to all-Internet Protocol (IP) frameworks, the NRSC's efforts are pivotal in addressing these transformative changes. By uniting the communications industry under a single voice, the NRSC effectively advocates for shared goals and solutions to common challenges.

America's communications networks are more than infrastructure—they are the backbone of our emergency response, commerce, and daily connections. The NRSC's unwavering dedication ensures these networks remain a source of national pride, supporting the strength, security, and resilience of the nation for generations to come.



ROBIN HOWARD
Brightspeed
NRSC CO-CHAIR



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Executive Summary

About the NRSC

The Alliance for Telecommunications Industry Solutions (ATIS) Network Reliability Steering Committee (NRSC) addresses network reliability improvement opportunities of service providers and vendors, in a noncompetitive environment, and allows participants to develop standards, technical requirements, technical reports, bulletins, Best Practices, and reports on the health of the nation's communications networks. The NRSC also coordinates industry improvements in network reliability through outage analysis. The mission statement[1] of the NRSC is:

The NRSC strives to improve network reliability by providing timely consensus-based technical and operational expert guidance to all segments of the public communications industry.

As a trusted expert, the NRSC addresses network reliability improvement opportunities in an open, noncompetitive environment. The NRSC advises the communications industry through developing and issuing standards, technical requirements, technical reports, bulletins, Best Practices, and annual reports.

The NRSC accomplishes this through:

- *identifying potential network reliability issues through an opportunity evaluation process,*
- *establishing subcommittees that address network reliability issues, and conducting special studies that may lead to industry recommendations and/or the development of Best Practices,*
- *developing industry feedback, both formal and informal, to the FCC on network reliability,*
- *providing industry feedback to the FCC on the Network Outage Reporting System (NORS) and the Disaster Information Reporting System (DIRS) and providing an opportunity for the public to be informed on network outages and ongoing efforts to resolve network reliability concerns.*

This Operational Report covers the period of 2023 through 2024. A brief history of the NRSC is provided in the *Introduction* of this report.

Changing Regulatory Environment & Changing Industry

The previous two Operational Reports (2019-2020 and 2021-2022) cited an increased focus on issues related to network reliability and resiliency and to the obligation of industry to report communications outages.

This focused attention continues to increase year-over-year, with numerous regulatory measures being enacted and industry responding to various high-profile service interruptions. Continuity of emergency services, information sharing, cybersecurity, and the continued move to an all-IP network have received the bulk of attention, although reliability of legacy networks remains a critical piece of the equation.

988

On July 21, 2023, the FCC released Report and Order FCC-23-57, titled *Ensuring the Reliability and Resiliency of the 988 Suicide & Crisis Lifeline* (PS Docket No. 23-5, PS Docket No. 15-80, WC Docket No. 18-336). This order modified Part 4 rules by requiring 988 service providers to report significant outages through the Network Outage Reporting System (NORS). It also established mandatory notification requirements for 988 crisis centers affected by service outages, ensuring that key stakeholders are informed and can take measures as needed.

Outage Reporting and Data Sharing

On August 29, 2023, the FCC issued FCC-23-71, titled *Wireless Resilient Networks Order on Reconsideration* (PS Docket No. 21-346, PS Docket No. 15-80, ET Docket No. 04-35). This order reaffirmed and clarified aspects of the Mandatory Disaster Response Initiative (MDRI), initially established in 2022. The MDRI requires wireless service providers to implement specific measures to strengthen network resilience during disasters. This ensures that communication services remain available for emergency responders and the public, even in adverse conditions. The order also addressed concerns from industry stakeholders while maintaining the FCC's focus on bolstering communication infrastructure during crises.

[1] This mission statement can be found on NRSC's webpage at < https://www.atis.org/committees-forums/nrsc/nrsc-mission/ >

On August 15, 2024, the FCC released FCC-24-73, titled *Resilient Networks; Amendments to Part 4 of the Commission's Rules Concerning Disruptions to Communications* (PS Docket No. 21-346, PS Docket No. 15-80, ET Docket No. 04-35). This order introduced additional amendments to Part 4 to improve outage reporting requirements. These updates are designed to provide more timely and actionable information to public safety entities, ensuring better coordination between service providers, emergency response agencies, and government regulators. The changes focus on enhancing network reliability and improving response efforts during large-scale disruptions.

These orders reflect the FCC's ongoing commitment to improving network resilience, strengthening emergency communications, and ensuring that critical services like 988 and disaster response networks remain operational during disruptions.

Highlights

During the 2023 to 2024 timeframe, the NRSC convened and reviewed and provided comments for high profile regulatory filings. Along with these initiatives, the NRSC also had three standing Subcommittees. The covered topics included:

Subcommittees

- Best Practices Subcommittee
- Outage Reporting Advisory Subcommittee
- Regulatory Subcommittee

NRSC Initiatives, Studies, and Filings

- Completed and Ongoing Initiatives:
  - Petition for Reconsideration on NORS/DIRS rules, seeking outage reporting waivers for special facilities and 9-1-1/988.
  - Petition for Reconsideration on NORS/DIRS, advocating balanced reporting requirements.
  - Reply Comments opposing broadband Title II reclassification mandates on Best Practices and performance reporting

- Filings[2]
  - *Reply to Opposition* in response to the Opposition of APCO International submitted by the Association of Public- Safety Communications Officials, International (APCO) in response to ATIS' Petition for Clarification and/or Reconsideration, July 3, 2024
  - *Letter* to Public Safety and Homeland Security Bureau regarding NRSC support for CSRIC Best Practices, May 23, 2024
  - *Comments* in response to the FCC Second FNPRM related to NORS and DIRS reporting obligations, May 13, 2024
  - *Petition for Clarification and/or Reconsideration* of the FCC Second Report and Order on NORS and DIRS reporting obligations, May 10, 2024
  - *Reply Comments* in response to FCC NPRM related to the application of the Network Outage Reporting System (NORS), January 17, 2024
  - *Ex Parte notification* on behalf of ATIS NRSC providing input on the California Major Service Interruption Reporting Portal, October 13, 2023

[2] Filings can be found on ATIS' Legal and Public Policy webpage at < https://www.atis.org/01_legal/public-policy/ >

- *Comments* in response to FCC rulemaking on advancing the transition to NG911, August 9, 2023

- *Comments* in support of Petition for Rulemaking regarding PSAP Outage Notifications, June 26, 2023

- *Reply Comments* in response to FCC rulemaking on the National 988 Suicide and Crisis Lifeline, June 6, 2023

- *Comments* response to FCC rulemaking on the National 988 Suicide and Crisis Lifeline, May 8, 2023

Publications

- NRSC Bulletins/Reports:
  - ATIS-0100071, *Network Reliability Steering Committee 2021-2022 Operational Report*

1 Introduction

1.1 History of the NRSC

1.1.1 Several Catastrophic Outage Events

From 1988 through the early 1990s, the United States communications industry experienced several network outages that impacted a large number of subscribers. Beginning with the "Great Hinsdale Fire" of 1988, through several Signaling Transfer Point (STP) outages in 1991, the nation increased its focus on the reliability of its public networks.

1.1.2 The Network Reliability Council is Established

In November 1991, the Network Reliability Council (NRC) was established by the FCC to bring together telecommunications industry leaders and telecommunications experts from academic and consumer organizations to explore and recommend measures to enhance network reliability.[3]

1.1.3 The FCC Mandates Outage Reporting

In April 1992, the FCC required the reporting of outages by exchange and interexchange service providers. For an event to be reportable, it had to last 30 minutes or more and potentially affect at least 50,000 customers.[4] The industry-led National Communications System (NCS) afterward recommended that the reporting criteria be lowered to 30,000 customers. Another NRC recommendation was to report all outages affecting 9-1-1 emergency call centers, major airports, nuclear power plants, major military installations, and key government facilities. Carriers began reporting outage events using the lowered threshold criteria in June 1992. Because of the sensitive nature of some of the outage events (e.g., military installations), in May 1993, the NCS accepted the task of reporting such outages to the FCC. In August 1994, FCC outage reporting regulations were revised.[5] Most of the changes had already been accounted for by industry in their voluntary reporting of events that began in June 1992. Other major changes included the reporting of fire-related incidents potentially affecting 1,000 or more lines, and the requirement that final reports include root cause analysis and a review of how Best Practices could have prevented or mitigated the impact of such events.

1.1.4 The NRC Recommends the Formation of the NRSC

In its 1993 *Report to the Nation*, the NRC[6] recommended the formation of the NRSC, under the auspices of the ATIS, for the purpose of monitoring network reliability on an ongoing basis. As defined at that time, the NRSC's mission was to "analyze the industry's reporting of network outages to identify trends, distribute the results of its findings to industry, and where applicable, refer matters to appropriate industry forums for further resolution, in order to help ensure a continued high level of network reliability."[7]

1.1.5 The FCC Makes Changes in Outage Reporting

In 2005, FCC regulations regarding outage reporting were put in force.[8] These mandates can be summarized as having three major aspects: (a) expansion regarding who was required to report; (b) new reporting thresholds, timeframes, and concepts; and (c) limited access to the outage data due to confidential protection under the Freedom of Information Act (FOIA). Regarding the reporting expansion, in addition to wireline providers, the new requirements included wireless, satellite, paging, and cable telephony service providers. Changes in the thresholds and concepts include events that affect 900,000 user-minutes and events impacting DS3 facilities. Because of these criteria, the overall number of reportable events substantially increased. In 2012, the FCC expanded the outage reporting criteria and thresholds to include VoIP services.

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[3] Daugherty, H.T., Klein, W. J., "U.S. Network Reliability Issues and Major Outage Performance," *Proceedings: IEEE Symposium on Computers and Communications, 1995,* June 27-29, 1995, pp.114, 119.

[4] *FCC Report and Order, CC Docket No. 91-273*, Federal Communications Commission, Washington, D.C., adopted February 13, 1992, released February 27, 1992.

[5] *FCC Second Report and Order, CC Docket No. 91-273*, Federal Communications Commission, Washington, D.C., adopted July 14, 1994, released August 1, 1994.

[6] Due to subsequent re-charters under the name "Network Reliability and Interoperability Council (NRIC)", this first Council is sometimes referred to as "NRC-1".

[7] *Network Reliability: A Report to the Nation*, Network Reliability Council, June 1993. Section I, p. 6.

[8] *FCC Report and Order and Further Notice of Proposed Rulemaking*, ET Docket No. 04-35, adopted August 4, 2004, released August 19, 2004; E*rrata*, ET Docket No. 04–35, released September 3, 2004.

In July of 2016, the FCC published a Report and Order[9] updating several of the Part 4, Disruption to Communications, rules with some significant changes. The changes: better defined required reporting at airports; shortened the threshold for reporting Simplex conditions from 120 hours to 96 hours; increased the minimal reportable transport outage from DS-3 to OC-3; defined the methodology for determining Wireless Potential Users; implemented required reporting for Partial PSAP Outage Reporting; and applied Telecommunications Service Priority (TSP) as the way of defining Special Offices and Facilities.

1.1.6 Disaster Response

In the summer of 2017, the United States and its territories experienced several hurricanes that shattered not only communication networks, but also people's lives. Hurricane Maria, in particular, devastated the island of Puerto Rico in a manner that had not been seen since Hurricane Mitch of 1998. In the aftermath of Hurricane Maria, the Information and Communications Technology (ICT) industry came together in a powerful way to enable recovery from this traumatic event. Carriers aided each other in procuring and providing generators, security teams, food, water, and the necessities to survive while restoring communications infrastructure. Parts of the island were without power and critical infrastructure for months, and the FCC activation of DIRS lasted 182 days, the longest in the history of DIRS.

1.2 Factors Affecting Network Reliability

The NRSC has historically recognized that identifying and understanding the underlying causes of outage trends are important parts of learning from past experiences and preparing for future challenges as networks evolve. When evaluating negative or positive trends that affect network reliability, having standard analytical methodologies and trending schemas has proven to be a solid link to the past, while providing a bridge into the future. The NRSC works to identify the direct and root cause(s) associated with particular trends, evaluates these against existing Best Practices, and develops new Best Practices or recommends modifications to existing Best Practices when appropriate. Additionally, the NRSC will recommend the development of new (or modification of existing) cause code categories, review other completed studies, review internal company outage data, determine contributing factors, and review associated federal and state regulations.

2 Health of the Nation's Public Networks

The NRSC members possess a historic and unique perspective on network reliability. Nowhere else in the world do subject matter experts from competing companies come together regularly to analyze network outage data, reach consensus on the findings, and provide expert guidance on actionable solutions to enhance network reliability. This collaboration fosters high reliability for the nation's public networks, offers valuable expert advice, and maintains an ongoing, accurate view of the health of networks at a national level. The NRSC remains confident that the reliability of the nation's public network is unparalleled globally.

2.1 Introduction to Special Studies/Task Forces

While the NRSC did not convene any Special Studies or Task Forces during this reporting period, the primary purpose of these studies is to focus industry experts on network reliability concerns, investigate the root causes of national trends, identify the most effective Best Practices or other strategies to prevent or mitigate the impact of such events, and provide industry-wide guidance on the issue. The success of these teams relies on open dialogue, meaningful information sharing, and collaboration among industry participants on potentially sensitive matters. To safeguard the interests of participating companies and their critical infrastructure data, a Non-Disclosure Agreement (NDA) is in place between NRSC member companies.

Studies reflect the countless hours NRSC members dedicate to thorough analysis, documenting, and publishing publicly available findings and results. These efforts play a crucial role in offering expert industry insights and maintaining high network reliability across the United States.

[9] FCC 16-63, *Amendments to Part 4 of the Commission's Rules Concerning Disruptions to Communications, PS Docket No. 15-80, ET Docket No. 04-35, PS Docket No. 11-82, Report and Order, Further Notice of Proposed Rulemaking, and Order on Reconsideration,* available at < https://apps.fcc.gov/edocs_public/ >

2.2 Subcommittees

2.2.1 Best Practices Subcommittee

2.2.1.1 Background

The Best Practices Subcommittee, which is a standing NRSC Subcommittee, is charged with improving the quality of Best Practices and updating and/or expanding them as appropriate. The Subcommittee operates from the premise that Best Practices are voluntary, are *not* standards, *and implementation of any Best Practices should not be mandated*. Best Practices provide guidance, based on assembled industry expertise and experience, to improve network security, reliability, and resiliency. The applicability and possible implementation of any Best Practice by an organization is best determined by someone with expertise in both the topic of the Best Practice and the particulars of the organization itself.

ATIS manages one of the two Best Practices websites[10], with the FCC maintaining the other site[11]. The Subcommittee monitors the ATIS website and suggests enhancements as appropriate to improve its usefulness.

2.2.1.2 Best Practices Team Activity

2.2.1.2.1 NRSC Stresses the Need to Maintain and Update Best Practices

On May 31, 2023, the NRSC stressed the need to maintain and update industry Best Practices to ensure network security, reliability, and interoperability. Recognizing that outdated practices could lead to confusion and compliance issues, the NRSC highlighted the gap left by the CSRIC's reduced role in managing these practices. To address this, the NRSC proposed creating a formal mechanism for regular updates, involving all stakeholders, and recommended updates every two years or as needed. In the absence of such a system, they suggested decommissioning the FCC database and relying on the industry to manage Best Practices. The NRSC also offered to assist the FCC in establishing an alternative process to ensure these guidelines remain effective and relevant.

2.2.1.2.2 NRSC Drafts CSRIC VIII Best Practices

In December 2023, NRSC drafted a list of potential Best Practices based on reports generated by CSRIC VIII Working Groups. NRSC emphasizes that these Best Practices were generated by CSRIC and only formatted for the database by NRSC. In April 2024, NRSC opened an Issue entitled “Review and Inclusion of CSRIC VIII Generated Best Practices into the Best Practices Database”. In May 2024, NRSC sent a letter to Suzon Cameron (FCC), CSRIC DFO, regarding the potential BPs identified in CSRIC VIII WG reports. This letter was filed as: Letter to Public Safety and Homeland Security Bureau regarding NRSC support for CSRIC Best Practices. In September 2024, the FCC received approval to accept NRSC’s offer to update the FCC’s CSRIC BP database. As of the publication of this report, executing this agreement and this work effort are still underway.

2.2.1.2.3 NRSC Offers Support to CSRIC on Best Practices

On May 23, 2024, the NRSC offered its support to the FCC in maintaining and enhancing the Communications Security, Reliability, and Interoperability Council (CSRIC) Best Practices database. The NRSC proposed assisting with incorporating CSRIC VIII Best Practices into the Commission’s database, ensuring they are formatted for easy integration and made publicly available to benefit the telecommunications industry. Additionally, the NRSC offered to support future CSRIC Best Practices, such as those from CSRIC IX and beyond, by identifying missing elements, structuring practices for database compatibility, assigning keywords for clarity, and providing training for CSRIC Working Groups. These efforts aim to reduce administrative burdens on the FCC, accelerate the adoption of updated practices, and enhance the visibility and utility of CSRIC’s work. The NRSC emphasized that collaboration would strengthen the reliability and resiliency of telecommunications while encouraging industry engagement with CSRIC initiatives.

2.2.1.2.4 NRSC Continues Review of Critical Communication Checklists

[10] The ATIS Best Practices website is available at < https://bp.atis.org >

[11] The FCC Best Practices website is available at < https://opendata.fcc.gov/Public-Safety/CSRIC-Best-Practices/qb45-rw2t/data >

In 2024, the Subcommittee continued reviewing ATIS-0100019(2022-01), *NRSC Emergency Preparedness and Recovery Checklist* as well as ATIS-0100018(2021-02), *NRSC Pandemic Checklist*, to ensure there are no duplicates of existing Best Practices. A detailed review is also underway to identify which new Best Practices should be proposed to CSRIC for inclusion in the Best Practice database.

2.2.1.3 Conclusion

The existence of an FCC Advisory Committee, such as CSRIC, which continually develops and refines Best Practices—along with their frequent citation in both industry and government documents—underscores the immense value of this collective industry knowledge and its impact on enhancing network security, reliability, and resiliency. The strength of synchronized Best Practice databases lies in both the expertise that shapes them and the voluntary nature of their implementation, allowing users to leverage their guidance while maintaining the flexibility to apply them as needed. The Best Practice Subcommittee will continue collaborating with the FCC to ensure the consistency and usability of this critical resource while updating supplementary materials, such as checklists, to reflect new insights and experiences. Additionally, the NRSC will work with the FCC to explore ways to incorporate Best Practices developed by external groups into the Best Practices database.

2.2.2 Regulatory Subcommittee

2.2.2.1 Background

The ATIS NRSC Regulatory Subcommittee addresses and responds to network reliability, resiliency, and outage reporting related regulatory activity. To accomplish this work, the Subcommittee monitors, reviews, and responds to various local, state, and federal regulatory activities. Where appropriate, the NRSC develops and files comments. Refer to the *NRSC Initiatives, Studies, and Filings* section of this report for a list of these filings.

2.2.2.2 Regulatory Team Activity

The Regulatory Subcommittee met consistently to establish and refine the issues for the NRSC membership and provide a forum for members to contribute to the comments on each of the Rulemaking procedures that were provided on behalf of ATIS NRSC. The comments provided to the FCC are available on the ATIS and FCC websites.

The Subcommittee also monitored and kept membership appraised of significant regulatory activity from state legislatures and commissions.

2.2.2.2.1 NRSC comments and reply comments on FCC Notice of Proposed Rulemaking (NPRM) Ensuring the Reliability and Resiliency of the 988 Suicide & Crisis Lifeline

In May and June 2023, the NRSC filed comments in response to this NPRM. These filings highlighted the differences in call flows for 911 versus 988, as well as the lack of data available regarding 988 outages (due to only one large-scale 988 outage having occurred prior to the NPRM). The NRSC suggested that the FCC not implement its suggested 988 certification, notification, and reporting requirements, without further investigation. Should the FCC move forward with these proposed changes, the NRSC proposed changes to the NORS platform to facilitate reporting.

2.2.2.2.2 NRSC comments on Competitive Carriers Association (CCA) Petition for Reconsideration for FCC Second R&O Amendments to Part 4 Rules, Improving 911 Reliability

In June 2023, the NRSC filed comments in support of this petition from CCA.

2.2.2.2.3 Joint comments from ESIF and NRSC in Response to the FCC NPRM on Facilitating Implementation of Next Generation 911 Services (NG911)

In August 2023, the NRSC and ESIF filed joint filed comments with several recommendations for the Commission's consideration.

2.2.2.2.4 NRSC files an ex parte notification on behalf of ATIS NRSC providing input on the California Major Service Interruption Reporting Portal

On October 13, 2023, the NRSC proposed key recommendations to enhance the California Major Service Interruption Reporting Portal. These recommendations included simplifying the input process for location information by enabling filers to select an "Other" option and enter data manually when necessary. Additionally, for multi-site outages, the NRSC recommended adding a checkbox to indicate "Multi-Site" and a text box for

explanation, along with a drop-down menu for consistent data analysis. To reduce errors and burdens, the NRSC suggested restoring the ability to save incomplete reports, allowing information from earlier reports to carry over into subsequent filings. NRSC also recommended enabling revisions to finalized reports, aligning the portal with the FCC's Network Outage Reporting System (NORS) standards. These recommendations aimed to streamline processes, enhance accuracy, and reduce administrative challenges for telecommunications service providers, ensuring a more effective and user-friendly outage reporting system.

2.2.2.2.5 NRSC reply comments in response to FCC NPRM related to the application of the Network Outage Reporting System (NORS)

In January 2024, the NRSC submitted reply comments, concurring with those who argue that no new outage reporting requirements are necessary. The NRSC also opposed proposals to mandate the use of industry Best Practices and disagreed with recommendations for the Commission to require reporting on speed, latency, or packet loss.

2.2.2.2.6 NRSC files a petition for clarification and/or reconsideration of the FCC Second Report and Order on NORS and DIRS reporting obligations

In May 2024, the NRSC filed a petition for clarification and/or reconsideration of the FCC Second Report and Order on NORS and DIRS reporting obligations. In this filing, NRSC requested the Commission to: (1) clarify the application of its waiver of NORS filings during DIRS activations to pending and ongoing NORS filings and to the Commission's Public Safety Answering Points (PSAPs) and 988 Suicide & Crisis Lifeline notification requirements, and to clarify that the waiver applies to both DIRS and DIRS-Lite activations; and (2) reconsider its requirements that the final DIRS report includes the estimated dates by which unresolved issues regarding providers' equipment and issues shall be resolved and these reports be filed within 24 hours of DIRS deactivation.

2.2.2.2.7 NRSC files comments in response to the FCC Second FNPRM related to NORS and DIRS reporting obligations

In May 2024, the NRSC filed comments in response to this FNPRM. These comments oppose the requirement of Broadband Internet Access Service (BIAS) to submit outage reports in NORS and to participate in DIRS. To the extent that NORS reporting is required for BIAS providers, the NRSC provided recommendations regarding the definition of BIAS "outages". The NRSC also opposes requiring providers to submit "after action" reports upon deactivation of DIRS. To the extent that these reports are required, the NRSC highlighted several recommendations for the Commission's consideration.

2.2.2.2.8 NRSC files a reply to opposition in response to the opposition of APCO International submitted by the Association of Public-Safety Communications Officials, International (APCO) in response to ATIS' Petition for Clarification and/or Reconsideration.

In July 2024, the NRSC filed a reply to opposition of APCO International. In this filing, NRSC reiterated the points made in its original petition for clarification and/or reconsideration on the FCC Second Report and Order on NORS and DIRS reporting obligations. ATIS disagrees with APCO that the Commission "lacks notice from an APA perspective" to grant ATIS' request for clarification regarding notifications to PSAPs and Emergency Communications Centers (ECC). Additionally, ATIS urged the Commission to reject the APCO opposition and grant its petition in its entirety. NRSC highlighted that the recommendations in the petition would reduce burdens for DIRS filers during emergency conditions and foster reliability by allowing service providers to focus on service restoration, rather than on reporting obligations

2.2.2.3 Conclusion

The Regulatory Subcommittee provides a platform to address and respond to network reliability, resiliency, and outage reporting related regulatory activity. The Subcommittee monitors, reviews, and responds to various local,

state, and federal regulatory activities. Where appropriate, the NRSC has developed and filed these comments or provided feedback to regulatory bodies.

2.2.3 Outage Reporting Advisory Subcommittee (ORAS)

2.2.3.1 Background

The NRSC established the Outage Reporting Advisory Subcommittee (ORAS) to review issues associated with reporting communication service disruptions pursuant to Part 4 of the FCC's rules. The ORAS was formed as a standing subcommittee that utilizes the experience and expertise of its members to improve the value, accuracy, and consistency of outage data submitted to the FCC, and since its establishment has expanded its role to address disaster information provided to the FCC on a voluntary basis. The ORAS works with the FCC to maintain a mutual understanding of the needs and expectations regarding submitted information, identifies process and system improvement opportunities, and develops appropriate recommendations, from the users' perspective, regarding enhancement of system interfaces, processes, and documentation.

2.2.3.2 NRSC ORAS Team Activity

2.2.3.2.1 NORS and DIRS Information Sharing for State Agencies Accessing Data

In 2023, the NRSC inquired whether the FCC would make available the entities that are able to access NORS and DIRS data. The FCC has since provided an email address (nors_dirs_information_sharing@fcc.gov) to gain updates to this information.

2.2.3.2.2 Determine Best Methods to Normalize Network Growth

In June 2023, the NRSC concluded an initiative aimed at addressing the FCC's suggestion to explore methods of normalizing network outage data in the context of network growth, particularly for broadband and OC3 networks. Initially prompted during a November 2021 meeting with the FCC, the NRSC committed to examining publicly available data to support this effort. Despite these efforts, the lack of consistent, publicly accessible network growth metrics and feedback led to the closure of this issue without producing a formal deliverable or work output. The NRSC's decision reflects its dedication to aligning resources with actionable outcomes while maintaining a focus on interoperability, reliability, and environmental sustainability across the telecommunications sector.

2.2.3.2.3 Review and Suggest Revisions to NORS Manual and Glossary

The subject matter experts (SMEs) in the ORAS reviewed NORS and DIRS documentation and provided suggested edits to the FCC for consideration. Several edits related to updated reporting for OC3 metrics and submarine cable outage reporting have recently been incorporated at the request of NRSC.

2.2.3.3 Conclusion

Through close examination of outage trends and reporting requirements, the ORAS seeks methods to ensure service providers and other stakeholders have well-defined guidance from the FCC. This guidance leads to a clearer understanding of patterns in outages, that assists both industry and regulators in identifying issues to be addressed in order to improve overall network reliability.

3 Report Conclusion

From 2023 to 2024, the NRSC has been deeply engaged in researching and providing expert guidance on a wide range of network reliability challenges while actively addressing FCC concerns related to network events. The NRSC remains dedicated to collaborating with the FCC to refine and improve outage reporting procedures, ensuring more accurate and actionable data collection. Additionally, it serves as a vital forum for fostering cooperative efforts between industry and government, promoting the exchange of critical insights and best practices. Through the unwavering commitment of its member companies, the NRSC has played a pivotal role in enhancing the resiliency and reliability of the nation's communications networks, reinforcing their ability to withstand disruptions and ultimately delivering widespread benefits to consumers, businesses, and emergency responders alike.

3.1 For the Common Good

The NRSC embodies the spirit of service within the communications industry, bringing together companies that, while fierce competitors in the marketplace, share a common commitment to improving network reliability. Through this unique collaboration, industry leaders set aside competitive interests to focus on strengthening the resilience, security, and performance of the nation's communications infrastructure. By working together, NRSC members drive meaningful advancements that benefit not only their own organizations but also consumers, businesses, emergency responders, and government agencies that rely on seamless and dependable connectivity. This dedication to collective problem-solving and industry-wide progress is one of the NRSC's most significant contributions, reinforcing its role as a driving force for the common good.

3.2 Future Plans

As the NRSC remains committed to enhancing the reliability and resiliency of today's networks, it also maintains a forward-looking approach to the industry's evolving landscape. Significant effort has been dedicated to developing a comprehensive model of an all-IP network, recognizing that the expertise gained from managing current networks will be crucial in addressing reliability challenges and outage reporting in this next-generation infrastructure. The NRSC's unique framework of industry collaboration, combined with its deep understanding of future network architectures, positions it as a vital resource during this period of technological transformation. Additionally, the NRSC continues to welcome input on emerging topics and encourages participation from both existing and new communications providers, fostering a cooperative environment that supports ongoing advancements in network reliability.

4 Participation

4.1 Participating NRSC Member Companies (2023-2024)

AQSACOM
AT&T
Brightspeed Connect Holding LLC
Charter Communications (Formerly Time Warner Cable)
Comcast
Comtech Telecommunications
Cox Communications
Ericsson
iconectiv
Intrado
Lumen
Nsight
Somos
T-Mobile
Telnyx
Verizon

4.2 NRSC Subcommittee Participants[12]

Best Practices Subcommittee (2023-2024)

Co-Chair: Andis Kalnins, Verizon
Co-Chair: Wendy Panella, AT&T

DeCuir, Jason	AT&T
Desiato, Robert	AT&T
DeVito, Victor	AT&T
Lawrence, Erik	AT&T
Rodocker, Susan	AT&T
Howard, Robin	Brightspeed Connect Holding LLC
Cummings, John	Charter Communications
Hall, Chad	Charter Communications
Nobles, Haleigh	Charter Communications

[12] The following lists provide meeting participation for the 2023-2024 period. NRSC engagement outside of meetings may not be captured.

Obasuyi, Thomson	Comcast
Smith, Thomas	Comcast
Browning, Rodney	Cox Communications
Peay, Mark	Cox Communications
Boyd, Mary	Intrado
Brown, Carolyn	Lumen
Wolf, Karen	Motorola Solutions
Bowie, Jason	T-Mobile
Wormsley, Becky	T-Mobile
Kalnins, Andis	Verizon
Desmond, Christopher	Verizon
Oberg, Chris	Verizon

Regulatory Subcommittee (2023-2024)

Co-Chair: Robin Howard, Brightspeed
Co-Chair: Paul Brown, Lumen

Berry, Darlene	AT&T
DeCuir, Jason	AT&T
Desiato, Robert	AT&T
DeVito, Victor	AT&T
Lawrence, Erik	AT&T
Magdic, Sandra	AT&T
Mosely, Richard	AT&T
Neinast, Mark	AT&T
Rodocker, Susan	AT&T
Tan, Mike	AT&T
Brown, Mary	Brightspeed Connect Holding LLC
Howard, Robin	Brightspeed Connect Holding LLC
Cummings, John	Charter Communications
Hall, Chad	Charter Communications
Nobles, Haleigh	Charter Communications
Collins, Kari	Comcast
Jordan, Beau	Comcast
Obasuyi, Thomson	Comcast
Smith, Thomas	Comcast
Lu, Ruobo	Comtech Telecommunications
Marshall, Roger	Comtech Telecommunications
Ornstein, Susan	Comtech Telecommunications
Browning, Rodney	Cox Communications
Peay, Mark	Cox Communications
Barclay, Steve	Ericsson
Boyd, Mary	Intrado
Brown, Carolyn	Lumen
Wright, Steven	Marconi Pacific
Nelson, Michael	Motorola Solutions

Wolf, Karen	Motorola Solutions
Green, Kevin	Somos
Ajayi, Solape	T-Mobile
Bowie, Jason	T-Mobile
Gormley, Andy	T-Mobile
Hagerson, Eric	T-Mobile
Wormsley, Becky	T-Mobile
Desmond, Christopher	Verizon
Kalnins, Andis	Verizon
Oberg, Chris	Verizon

Outage Reporting Advisory Subcommittee (2023-2024)

Co-Chair: Mike Tan, AT&T
Co-Chair: Becky Wormsley, T-Mobile

Berry, Darlene	AT&T
DeCuir, Jason	AT&T
Desiato, Robert	AT&T
DeVito, Victor	AT&T
Lawrence, Erik	AT&T
Neinast, Mark	AT&T
Cummings, John	Charter Communications
Hall, Chad	Charter Communications
Nobles, Haleigh	Charter Communications
Collins, Kari	Comcast
Obasuyi, Thomson	Comcast
Lu, Ruobo	Comtech Telecommunications
Peay, Mark	Cox Communications
Rubio, Ed	Cox Communications
Barclay, Steve	Ericsson
Boyd, Mary	Intrado
Brown, Carolyn	Lumen
Wright, Steven	Marconi Pacific
Wolf, Karen	Motorola Solutions
Wormsley, Becky	T-Mobile
Casem, David	Telnyx
Desmond, Christopher	Verizon
Oberg, Chris	Verizon

4.3 Companies/Organizations in Attendance at the 2023-2024 Public NRSC Quarterly Meetings

AT&T
BBT
Brightspeed Connect Holding LLC
Buckeye Broadband
Charter Communications
Cincinnati Bell
Columbus Networks

Comcast
Comtech Telecommunications
Cox Communications
East Kentucky Network
Ericsson
eXclusive Services
Fastwyre Broadband
FCC
GCI
Golden West Communications
Intrado
Lafayette Utilities System
LTC Connect
Lumen
Marconi Pacific
Mississippi State Dept. of Health
Mountain Telephone
MTA
Nex-Tech
NGA
NortheastTel
Northwest Communications Cooperative
Nsight
Polar Communications
Range
Shawnee Health
Smart City
Somos
Spectrum
Star Communications
Swiftel Communications
T-Mobile USA
TDS Telecom
Union Wireless
Verizon
West
Whidbey Telecom
Windstream
Yelcot